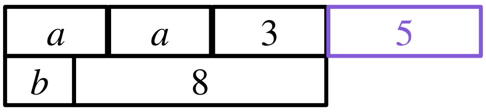
Name: ______________________

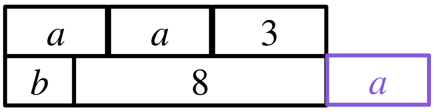
Worksheet

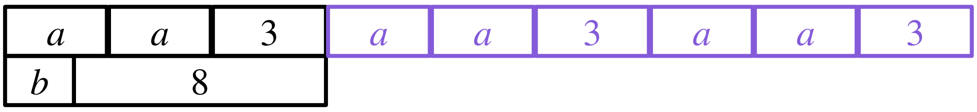
Securing understanding of equality

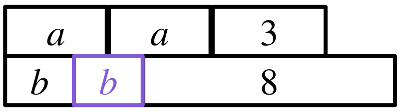
Task A: Maintaining equality in bar models

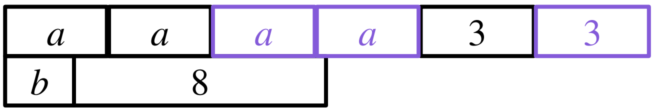
1) Complete each bar model so it maintains equality, given that $2a + 3 = b + 8$

a)


b)


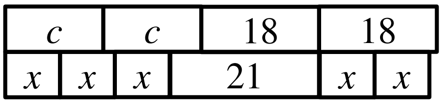
c)


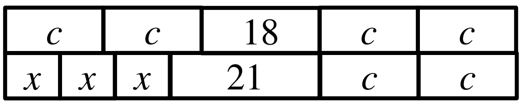
d)


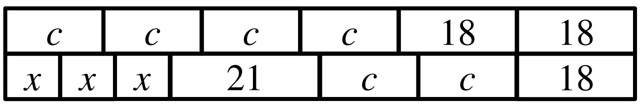
e) e.g.


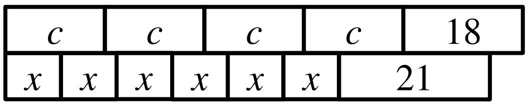
2) The bar models are not drawn to scale. Given that $2c + 18 = 3x + 21$

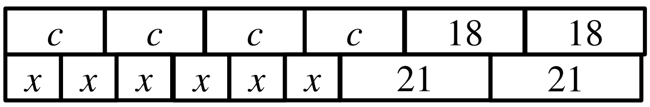
Which bar models will maintain equality, and should be the same length?
Explain your reasoning.

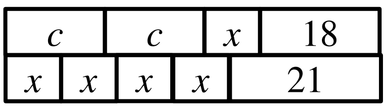
a)


b)


c)


d)


e)


f)


Task B: Maintaining equality in equations

1) Fill in the blanks in these equations so they maintain equality.

a)

$$a = b$$
$$a + 8 = \underline{\hspace{3cm}}$$
$$a - 2 = \underline{\hspace{3cm}}$$

b)

$$a = b$$
$$a\underline{\hspace{2.5cm}} = b + r + 5$$
$$a\underline{\hspace{2.5cm}} = b + r + 3$$

c)

$$a = b$$
$$\underline{\hspace{3cm}} = 3b$$

d)

$$3x + 4 = y + 1$$
$$3x + 4 - x = y + 1\ \underline{\hspace{2cm}}$$

e)

$$3x + 4 = y + 1$$
$$\underline{\hspace{3cm}} = 8(y + 1)$$

f)

$$3x + 4 = y + 1$$
$$3x + 4 + (-5) = \underline{\hspace{3cm}}$$

g)

$$p + q = 10$$
$$\frac{p + q}{5} = \underline{\hspace{3cm}}$$

h)

$$p + q = 10$$
$$\underline{\hspace{3cm}} = 30$$

i)

$$p + q = 10$$
$$p + q + p + q = \underline{\hspace{3cm}}$$